

Supplementary Material for “A Two-Stage Reconfigurable Intelligent Surface Architecture for Decoupled M-FSK Modulation and Beamforming in 28-GHz Bistatic Sensing”

Equation, figure and table numbers without the prefix S refer to the main paper.

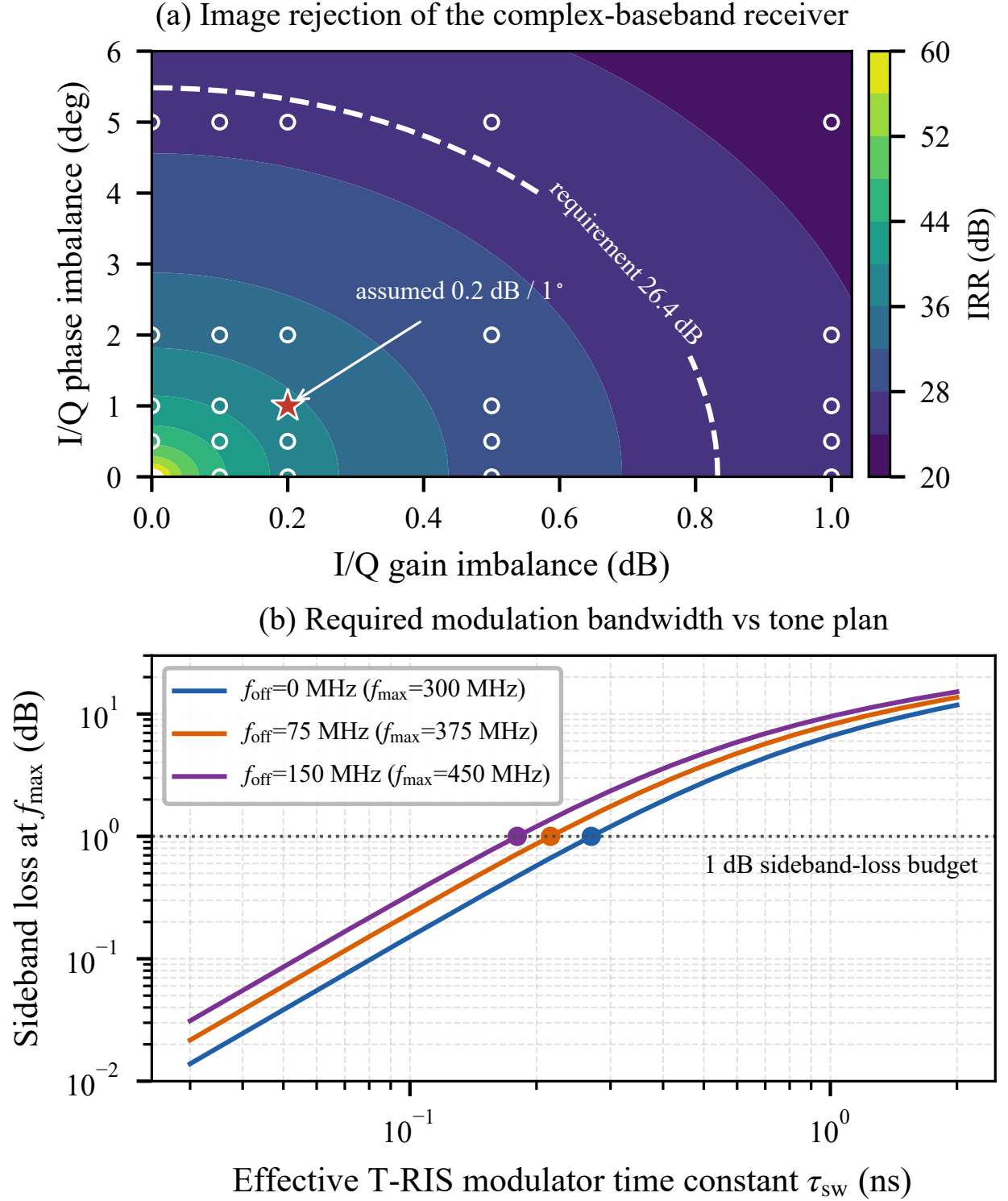


Fig. S1: (a) Image-rejection ratio (31) versus I/Q gain and phase imbalance; circles are time-domain correlator simulations, the dashed contour is the 26.4 dB requirement and the star the assumed imbalance. (b) Sideband loss at the highest tone versus the effective T-RIS modulator time constant for three tone plans; dots mark the 1 dB limit.

TABLE S1: Tone-plan options: RF guard band versus required T-RIS modulation bandwidth

f_{off} (MHz)	Tones (MHz)	Image guard (MHz)	$\tau_{\text{sw}}^{\text{max}}$ (ns)	Max squint (deg, % HPBW)	ΔR_{path} (m)	$\sqrt{\text{CRB}_L}$ (cm)
0	25–300	50 (0.18 %)	0.27	1.04 (8.2 %)	1.00	1.87
75	100–375	200 (0.71 %)	0.22	1.29 (10.1 %)	1.00	1.87
150	175–450	350 (1.25 %)	0.18	1.53 (12.1 %)	1.00	1.87

$\tau_{\text{sw}}^{\text{max}}$: largest effective modulator time constant keeping the sideband loss at f_{max} within 1 dB. Squint at $\theta_1 = 60^\circ$. Resolution and CRB are independent of f_{off} ; operating point $\text{SNR}_1 = -14.5$ dB.

TABLE S2: Multi-Tone Beam Coherence of the Two-Stage R-RIS ($N_x = 16$, $M = 12$, $f_{m,M} = 300$ MHz)

θ_1 (deg)	Design freq.	HPBW (deg)	Max squint (deg, % HPBW)	Worst (dB)	Aggr. (dB)	CRB (+%)	Bias (mm)
30	f_c	7.3	0.35 (4.8)	0.026	0.010	0.13	0.00
30	centre	7.3	0.16 (2.2)	0.005	0.002	0.04	0.00
60	f_c	12.7	1.04 (8.2)	0.079	0.029	0.40	0.00
60	centre	12.7	0.49 (3.9)	0.016	0.006	0.13	0.00
75	f_c	24.5	2.12 (8.7)	0.098	0.037	0.50	0.00
75	centre	24.5	1.09 (4.4)	0.021	0.008	0.17	0.00

Worst/Aggr.: loss toward the target of the worst tone and of the mean tone power, relative to the design frequency. CRB: increase of (39) from the tone-dependent gains. Bias: noise-free bias of the ML path estimate. A single-stage STM-RIS re-steered per symbol has zero squint but reloads M phase maps per sweep.

TABLE S3: Positioning Versus Receive Aperture ($\beta = 60^\circ$, Rx–Target Distance 8 m)

N_{rx}	D (cm)	G_{rx} (dBi)	$2D^2/\lambda$ (m)	$\sqrt{\text{CRB}_L}$ (cm)	$\sqrt{\text{CRB}_\theta}$ (deg)	$\sqrt{\text{CRB}_{\text{pos}}}$ (cm)	RMSE focused	RMSE far field
4	1.6	12	0.0	3.74	1.1048	17.99	19.00	19.00
8	3.7	15	0.3	2.65	0.3812	6.39	6.71	6.70
16	8.0	18	1.2	1.87	0.1340	2.49	2.46	2.46
32	16.6	21	5.1	1.32	0.0473	1.17	1.15	1.15
64	33.7	24	21.2	0.94	0.0167	0.68	0.70	0.72
128	68.0	27	86.3	0.66	0.0059	0.45	0.45	14.31

Half-wavelength spacing; G_{rx} and SNR_1 scale with N_{rx} from the 16-element values of Table I. RMSE in cm (Monte Carlo); “focused” scans with the Rx–target distance known from the path estimate.

TABLE S4: Transmit-Side DC Power Budget, Identical Accounting ($N = 256$)

Item	W
<i>Two-stage RIS (PIN, 2 mW/element/layer, $E_{\text{sw}} = P_{\text{bias}}\tau_{\text{sw}}$)</i>	
CW source (PA, 20 T-RIS PIN bias)	0.512
R-RIS bias	0.512
T-RIS switching	0.045
Controller, DDS, clock	0.150
Total	1.719
<i>AESA, 5 mW / 15 mW per channel</i>	
T/R channels	1.280
LO distribution, main amplifier	0.870
Controller	0.150
Total (5 / 15 mW)	2.300 / 4.860
Common bistatic receiver (excluded)	0.630